

Juan David Munoz Bolanos

Optical Systems Engineer

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PROFESSIONAL SUMMARY

Optical Systems Engineer with a Ph.D. in adaptive optics and a track record of end-to-end precision instrument development: from interferometric wavefront sensing through C++/Python/LabVIEW integration and peer-reviewed validation. Deep expertise in optical wave-field simulation and Zernike-based aberration modeling; familiar with ZEMAX OpticStudio for tolerancing. ZEISS Summer School alumnus. Available May 2026.

TECHNICAL COMPETENCIES

Wavefront Sensing & Interferometric Metrology

Designed and deployed F-SHARP interferometry and batched-DASH wavefront optimization for sub-wavelength aberration correction. Novel spatial-shift algorithm ("3N method") enabling high-accuracy phase retrieval in diffraction-limited systems.

Optical Simulation & Modeling

Deep knowledge of wave-field propagation modeling and Zernike polynomial decomposition, implemented in custom Python simulation frameworks. Familiar with ZEMAX OpticStudio for optical tolerancing and aberration sensitivity analysis.

Opto-Mechatronic Integration & Automation

Full hardware-software integration of adaptive optics systems (deformable mirrors, SLMs, galvo scanners, scientific cameras) via C++/Python/LabVIEW. Automated calibration and alignment routines ensuring repeatable, high-precision performance.

Machine Vision & Industrial Inspection

Industrial deployment of Halcon and OpenCV pipelines for high-speed automated inspection and defect detection. Custom illumination design (bright-field, dark-field, ring-light) for challenging surface-inspection tasks.

PROFESSIONAL EXPERIENCE

Medical University of Innsbruck

Optical Systems Engineer (PhD Researcher)

Innsbruck, Austria

Jan 2022 – Present

- **Wavefront Sensing System (ACS Photonics '25 / Biomed. Opt. Expr. '23):** Designed and built from scratch a fully automated interferometric wavefront sensing platform. Integrated 63-actuator Deformable Phase Plates and TI Phase Light Modulators via a custom C++/Python/LabVIEW stack. Achieved diffraction-limited resolution in multi-photon deep-tissue imaging, validated through two peer-reviewed publications.
- **Interferometry & Phase Retrieval:** Implemented F-SHARP wavefront interferometry and developed a novel spatial-shift estimation algorithm ("3N method") enabling high-fidelity Zernike decomposition of experimental aberrations. Benchmarked against custom Python wave-propagation simulations.
- **Optical Simulation & Tolerancing:** Built custom Python wave-field simulation frameworks using Zernike decomposition to predict and characterise aberration sensitivity throughout the optical path. Familiar with ZEMAX OpticStudio for optical tolerancing support.
- **System Procurement & Qualification:** Specified and procured high-precision opto-mechanical components, DAQ electronics, and interferometric sensors. Validated incoming components against technical specifications before integration into live experiments.

Leibniz Institute of Photonic Technology (IPHT)

R&D Researcher – Photonics & Spectroscopy

Jena, Germany

Jun 2020 – Dec 2021

- **Dispersive Spectrometer Design (Applied Sciences '24):** Led optical layout, ZEMAX tolerancing, and construction of a 1064 nm fiber probe Raman spectrometer for clinical tissue applications. Handled detector alignment, wavelength calibration, and fiber coupling from prototype to published instrument.
- **Hyperspectral ML Classification:** Applied supervised machine learning to 2D hyperspectral Raman tissue images, achieving high-accuracy cancer vs. healthy cell separation. Accelerated experimental characterization loops significantly.

INTECOL S.A.S.

Junior Machine Vision System Engineer

Medellín, Colombia

Mar 2018 – Mar 2019

- **Industrial Vision Pipelines:** Deployed Halcon-based inspection systems on high-speed glass-bottle production lines. Engineered custom bright-field, dark-field, and ring-light illumination schemes; resolved label-verification and liquid-level failures. System reached full customer sign-off.
- **Hardware Specification:** Selected industrial cameras, lenses, and sensors; implemented and validated full optical inspection setups on live production lines.

EDUCATION

Medical University of Innsbruck

Ph.D. in Adaptive Optics & Precision Optical Metrology

Austria

Jan 2022 – Present

Friedrich Schiller University Jena

M.Sc. in Photonics

Germany

2019 – 2021

University of Cauca

B.Sc. in Physics Engineering

Colombia

2012 – 2018

PUBLICATIONS & AWARDS

- **Awards:** Competitively selected for ZEISS Summer School (precision optics engineering, Oberkochen). Awarded COLFUTURO Grant 2026 for top Colombian professionals.
- Juan David Munoz Bolanos et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* 12(1), 176–184 (2025). Integrated adaptive optics into a confocal Raman system; experimentally resolved and corrected phase aberrations for diffraction-limited deep-tissue imaging.
- Juan David Munoz Bolanos et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. *Applied Sciences*, 14(11), 4726 (2024). Full optical design, tolerancing, construction, and clinical validation of a custom dispersive fiber probe spectrometer.
- Sohmen M., Juan David Munoz Bolanos et al. Optofluidic adaptive optics in multi-photon microscopy. *Biomedical Optics Express* 14(4), 1562–1578 (2023). Integrated optofluidic tuneable lenses into a multiphoton microscope with custom Python/LabVIEW control for fast, aberration-free focusing.

LANGUAGES

English (Fluent / Professional),
German (B2 - Working Proficiency),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte

Head of Institute for Biomedical Physics

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Group leader for Spectroscopy Imaging

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Leibniz Institute of Photonic Technology